

Probabilistic Project Cost Modelling using Risk Evaluation

Group 9: Synoptix

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Declaration of AI use

AI tools have been used throughout this project in various ways, from helping identify relevant areas and methods to assisting with the coding. One of the key aids was with initial idea generation. AI also played a key role in aiding in coding various Bayesian networks. However, AI was *not* involved in writing this report.

1 Introduction

The demand for engineering solutions is rising, with the global engineering services market expected to reach \$1.74 trillion in 2025 [1]. The engineering software sector is expected to experience the largest growth at 20.3% per year from 2025 to 2030 [2]. As these industries expand, systems engineering firms play a crucial role in managing complex projects efficiently. Accurate cost estimation is fundamental to this through resource planning, risk management, and adherence to deadlines. Recent studies indicate that on average, IT projects run 45% over budget and 7% over time [3]. Cost prediction is challenging due to market fluctuations, regulatory changes, and unforeseen risks. Addressing these challenges is essential to prevent budget overruns, which can affect project objectives and stakeholder confidence.

The earliest approaches to cost estimation rely on deterministic methods and expert judgment. For example, Synoptix, who proposed this project, currently estimates costs with a fixed $\pm 10\%$ margin, limiting the adaptability to risks. However, these methods do not account for uncertainty or dependencies. As a result, they often fail to account for how interconnected risks, such as employee absences and supply chain delays, can compound and significantly impact costs [4]. Monte Carlo Simulations (MCSs), introduced in the 1940s, marked a major advancement by using probabilistic modelling to simulate thousands or even millions of scenarios based on risk likelihood and impact. This approach generates a distribution of outcomes, enabling you to identify optimistic and pessimistic cost bounds and deal with the uncertainty and variability of a problem [5]. However, it assumes independence between input variables, limiting its ability to capture interdependencies that are common in real-world projects [6].

To address this, Bayesian Networks (BNs) model relationships between risks and allow for the representation of dependencies, enabling cost estimations that reflect real-world scenarios. In particular, Static BNs (SBN) provide a snapshot of these relationships at a specific point in time, making them effective for analysing project risks in stable conditions [7]. Bayesian Hierarchical Models (BHMs) introduced a framework to address the complexities of uncertainty across multiple variables in a project. By capturing both local and global factors they enable a more comprehensive approach to cost estimation. This multilayered perspective further refines accuracy, especially in large-scale systems [8]. Finally, Dynamic Bayesian Networks (DBNs) explore temporal dependencies. This models how risks and their interactions evolve over time, making DBNs particularly valuable in multi-phase projects where conditions can shift significantly throughout the lifecycle [9].

This study assesses the strengths and limitations of each method to determine their suitability for different projects, offering a framework for companies to select the most appropriate model. The report begins with MCS as a foundational approach, incorporates dependencies through SBN, refines cost estimation with BHM, and adopts DBN to capture temporal and sequential relationships.

2 Monte Carlo Simulation Model

Monte Carlo Simulations (MCS) is a widely used probabilistic modelling technique for project cost estimation, achieved by generating possible cost scenarios. They provide a distribution of outcomes, offering insight into cost variability and risk exposure. This approach is especially valuable in industries where projects are prone to fluctuating costs, as it efficiently analyses schedule risks, making it a useful tool for project managers [10]. By modelling a range of potential outcomes, MCS helps mitigate financial risks and improves decision-making in project

management.

Phase	Base Cost	Risk Level	Engineer Type
1.1	280	0.4	Project Engineer
1.2	140	0.6	Project Engineer
1.2.1	360	0.8	Principal Systems Engineer
1.3	240	0.4	Principal Systems Engineer

Table 1: Sample of dataset provided by Synpotix

The dataset provided by Synoptix contains four key inputs. The four phases, each containing different numbered tasks. The Base cost, representing the initial estimated cost for each task. The Risk level which quantifies the uncertainty associated with each phase and is set at 0.2, 0.4, 0.6, or 0.8, reflecting the increasing degrees of variability. The risk level directly influences the standard deviation of cost projections, as defined by: $\sigma = \text{Base Cost} \times \text{Risk Level}$, where a higher risk level results in greater variability in cost projections. Finally, the engineer type, whether Project Engineer, Systems Engineer, or Principal Systems Engineer, is associated with an hourly cost of £70, £80, and £120, respectively. Table 1 provides a sample of this dataset.

2.1 Method

The MCS was implemented to estimate project costs under uncertainty using a log-normal distribution, which is particularly suited for modelling financial data where cost overruns are more likely than savings [11]. Empirical studies have shown that cost data in real-world projects often exhibit a right-skewed distribution, making the log-normal distribution a better fit than symmetrical alternatives [12] like the normal distribution. Additionally, the log-normal distribution ensures that all simulated costs remain positive, aligning with the fundamental nature of project expenditures, which cannot fall below zero [7].

The MCS was applied by defining the log-normal distribution for each phase, where the mean was set as the base cost and the standard deviation was determined by multiplying the base cost by the risk level. A total of 10,000 random samples were generated for each phase to simulate potential project outcomes. This sample size was chosen based on its widespread use in Monte Carlo analyses, which ensures a balance between computational efficiency and result accuracy [5].

2.2 Results

The log-normal distribution introduces a right skew, reflecting the greater likelihood of cost overruns than savings. This asymmetry is crucial for capturing real-world cost variations, where extreme overruns occur more frequently than large savings. The 25th, 50th, and 75th percentiles provide optimistic, median, and pessimistic estimates, respectively. For the log-normal distribution, these are £4,807.5, £6,506.4, and £8,957.1, while for the normal distribution, they are £4,769.3, £7,452.6, and £10,083.7. These results show that the log-normal distribution better captures extreme cost variability, particularly in the upper range, while the normal distribution assumes symmetry and underestimates the likelihood of significant overruns. Using percentiles ensures a robust approach by reducing the influence of outliers [13]. Figure 11 (in Appendix) further illustrates these differences.

Figure 1 illustrates cost variability across tasks using Monte Carlo simulations, with task colours corresponding to their associated risk levels. Tasks with higher base costs and risk levels, such

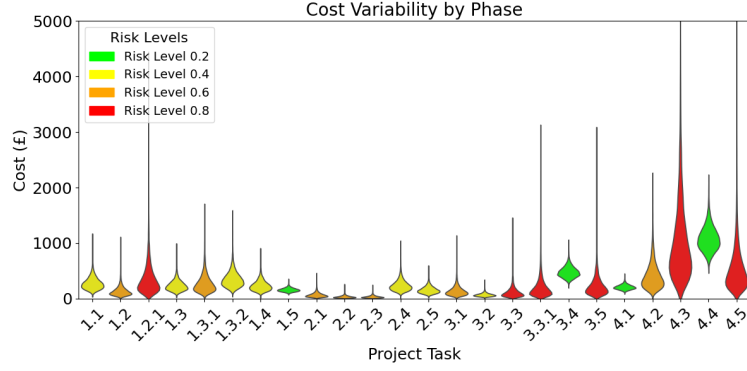


Figure 1: Violin plot showing cost variability across project tasks for risk levels 0.2, 0.4, 0.6, and 0.8. Tasks with higher risk levels exhibit greater cost variability.

as 4.3 and 4.4, exhibit wider distributions, indicating greater uncertainty and higher potential for cost overruns. In contrast, tasks with lower base costs, like 1.1 and 2.1, show narrow distributions, reflecting minimal variability. The asymmetric shapes of the violins, particularly for high-risk tasks, emphasise the right-skewed nature of the log-normal distribution, accurately capturing the likelihood of extreme cost overruns.

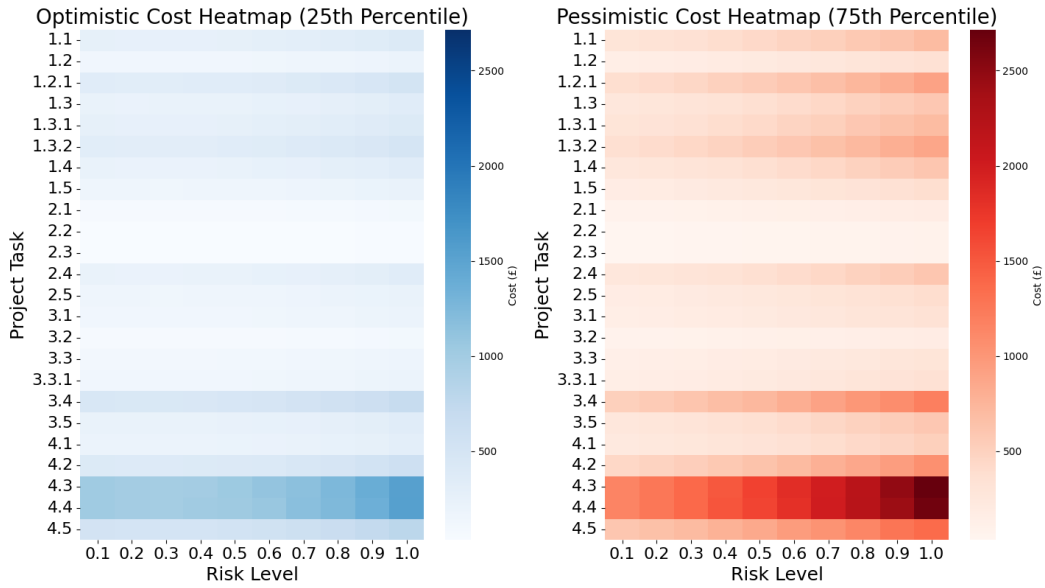


Figure 2: Heat maps showing optimistic (left) and pessimistic (right) project costs across risk levels and tasks. Optimistic costs increase gradually with risk, while pessimistic costs rise sharply, especially in later tasks, reflecting the log-normal distribution's right skew.

Figure 2 shows the variation in optimistic and pessimistic project costs across risk levels and tasks. The left heat map illustrates a gradual increase in optimistic costs, reflecting lower sensitivity to risk. In contrast, the right heat map reveals a sharp rise in pessimistic costs with higher risk levels, particularly in later phases. This disparity highlights the amplification of risks in pessimistic scenarios, consistent with the right-skewed nature of the log-normal distribution, which disproportionately increases costs under higher risks.

However, a key limitation of MCS is evident in these heat maps, they fail to account for task dependencies. This leads to isolated risk evaluations, which oversimplify the interconnected

nature of tasks in real-world projects. MCS quantifies cost variability well, but it struggles with modelling dependency-driven risks, limiting its use in complex projects.

3 Static Bayesian Network Model

A Static Bayesian Network (SBN) is a probabilistic graphical model that captures dependencies between tasks. It has been widely used to model complex systems with dependent variables. For example, SBNs have been applied in assessing national economic competitiveness, effectively modelling dependencies between economic indicators, innovation and investment factors to predict economic trends [14]. This demonstrates their ability to analyse relationships within structured data, making them well-suited for evaluating the relationship between risk and cost in project management.

3.1 Method

The SBN assigns each project task a discrete risk state: (Low, Medium, High and Very High) based on the risk level provided by Synoptix. Each task’s risk level is modelled as a node with edges representing conditional dependencies. This allows the model to capture chain reactions within a project: a high risk event in one task increases the likelihood of overrun in another, which would have otherwise been treated independently in MCS.

The nodes are connected based on a logical task sequence, with dependencies represented through Conditional Probability Distributions (CPDs). These CPDs define how the risk state of a child node is influenced by the risk state of its parent node. This relationship can be mathematically expressed using Bayes’ rule as follows:

$$P(R_c = r | R_p = s) = \frac{P(R_p = s | R_c = r) \cdot P(R_c = r)}{P(R_p = s)}, \quad (1)$$

where R_c and R_p are the risk states of the child and parent tasks, $P(R_c = r)$ is the initial probability determined by the tasks risk score and $P(R_p = s | R_c = r)$ quantifies how likely it is for the parent task to exhibit different risk states given that the child task is in state r .

Child node CPDs are stored as 4×4 matrices, with rows representing the child’s risk state and columns representing the parent’s risk state. This structure ensures that a parent node with Very High risk increases the risk of its child node. Conversely, if the parent’s risk is Low, the child node’s distribution stays near its baseline. This effect is evident in tasks 4.3 and 4.4, where task 4.3, classified as Very High risk, propagates increased risk to task 4.4, which would otherwise fall into the Low-risk category. Once the Bayesian Network is constructed, forward sampling generates 10,000 risk state samples per task. A risk factor map assigns levels: 0.8 for Low, 1.0 for Medium, 1.2 for High, and 1.4 for Very High [15]. These levels are applied to each task’s base cost to compute final costs.

3.2 Results

Figure 3 illustrates the probability of tasks falling into different risk states across project phases. Tasks in earlier phases (1 and 2) are primarily in Low and Medium risk categories, reflecting limited variability. In contrast, tasks in later phases (3 and 4) shift significantly towards Very High risk, with Phase 4 showing the highest concentration. This identifies Phase 4 as the main driver of overall cost uncertainty and where risk mitigation efforts should be focused.

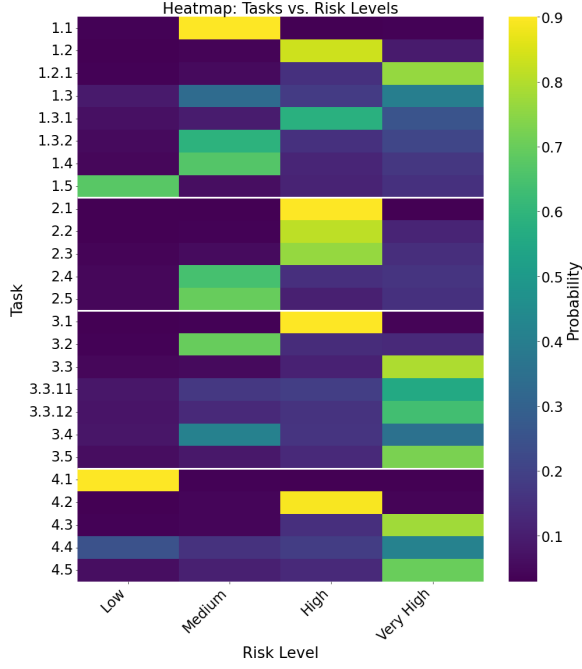


Figure 3: Heat map showing the probability distribution of risk states across project phases, with rows summing to 1. The white lines separate the phase boundaries.

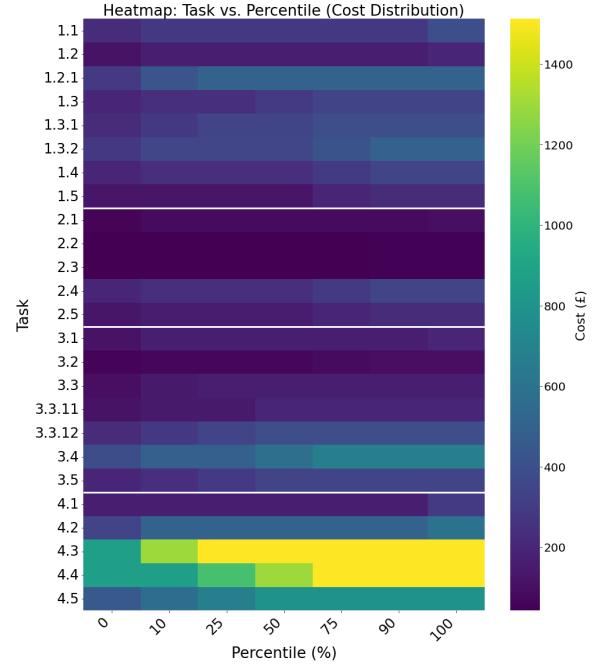


Figure 4: Heat map showing cost distribution across percentile increments. The white lines separate the phase boundaries.

Figure 4 presents the cost distribution of each task across different percentiles. It highlights task 4.3 as consistently having the highest costs across all percentiles. This indicates that 4.3 is prone to increased uncertainty, making it one of the biggest contributors to overall cost variability. Task 4.4 and then 4.5 also show elevated costs due to being the child nodes and having increased risk.

Since the SBN treats each phase independently, it is useful for single level probabilistic cost estimation. This makes it especially useful during early-stage project planning because of its computational efficiency and ability to model uncertainty without the need for a lot of data [16].

4 Bayesian Hierarchical Model

The Bayesian Hierarchical Model (BHM) is designed to analyse multilevel datasets by modelling variability at each level and sharing information between groups within the same level [17]. It has been applied in real-world cases where there are global and local datasets that require analysis for example, studying the effects of a drug on patients across different hospitals[18]. The BHM allows for comparison between hospitals while simultaneously comparing data for patients within a hospital. By combining prior knowledge with observed data, BHM creates improved estimates at each hierarchical levels. Since the data provided by Synoptix was limited for analysis, distributions for variables such as engineer costs were generated, enabling a broader range of data variation and more robust analysis than other models.

4.1 Method

This cost estimation framework follows a BHM, where costs at different levels, engineering roles, tasks, and the overall project are modelled probabilistically. At the first level, distributions are assigned to the base costs of engineers, capturing uncertainty in their hourly rates. Initial hourly costs (C_e) are £70 for Project Engineers (PE), £80 for Systems Engineers (SE), and £120 for Principal Systems Engineers (PSE). Variations in costs (C_a) are modelled using a normal distribution, $C_a \sim \mathcal{N}(90, 10^2)$, where £90 is the mean, representing the average of initial costs, and the standard deviation of £10 reflects role-based variability. The normal distribution is widely used in cost estimation due to its symmetrical properties, making it appropriate when modelling variations that are equally probable above and below the mean [19]. To ensure realistic values, a truncated normal distribution is applied with bounds calculated as $\text{Bounds} = C_e \pm \sigma_e$, resulting in £60–£80 for Project Engineers, £70–£90 for Systems Engineers, and £110–£130 for Principal Systems Engineers.

At the second level, task costs are adjusted based on the engineer’s costs, allowing cost variations to propagate through the model. The adjusted task cost is given by $T_a = C_t/C_e \times C_a$, where C_t is the task’s initial cost and T_a is the adjusted task cost.

The third level introduces risk levels, accounting for variability in task costs due to risk. A distribution is applied to improve sensitivity to risk scores: $R_a \sim \mathcal{N}(0.5, 0.13^2)$, where R_a is the adjusted risk level truncated to the range $[0.1, 1.0]$. The mean, 0.5, reflects the midpoint and the standard deviation, 0.13, captures the entire range, with higher risk scores corresponding to higher levels.

Finally, task costs under optimistic and pessimistic scenarios are modelled using a log-normal distribution, consistent with the rationale in the MCS section. This hierarchical approach ensures that uncertainty is consistently represented across all levels of project cost estimation.

4.2 Results

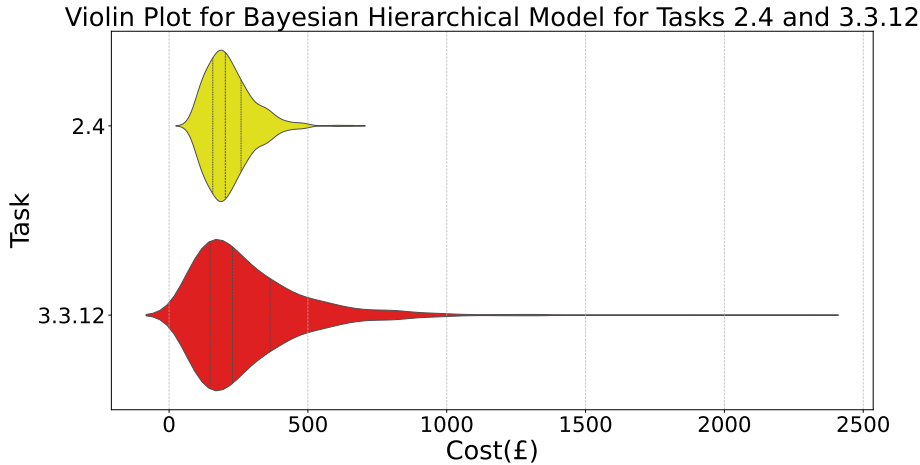


Figure 5: Violin Plot for Tasks 2.4 and 3.3.12 showing the variability of the cost distribution for each task. The left and right inner lines represent the optimistic and pessimistic costs, respectively, while the middle line represents T_a . These tasks were chosen at random to show the distribution of those tasks. The colour determines the risk category of the task.

Figure 5 presents a violin plot comparing the cost distributions for Tasks 2.4 and 3.3.12 within

the BHM because they share similar base costs. The results highlight the impact of risk levels on cost variability, demonstrating that risk plays a more significant role in shaping the distribution than the base engineering cost alone. Task 3.3.12 exhibits a notably wider distribution with a pronounced right tail, indicating a higher probability of extreme cost overruns. This is a direct consequence of its higher risk level, which amplifies cost fluctuations at Level 3 of the model. The adjusted risk level applied at this stage increases variance, leading to greater uncertainty in final cost estimates.

In contrast, Task 2.4 displays a more constrained cost distribution, with significantly less variability despite being assigned to a higher-cost engineering role (PSE). This stability can be attributed to its lower risk level, which limits the influence of the risk level at Level 3, resulting in a more predictable cost estimate. The comparison between these two tasks underscores the importance of incorporating probabilistic risk adjustments into project cost estimation. While engineering roles introduce baseline cost variability, it is the risk classification that ultimately dictates the degree of cost uncertainty. Consequently, tasks with high-risk levels require more robust contingency planning to mitigate the potential for cost overruns. These findings reinforce the necessity of risk-aware budgeting strategies when applying hierarchical Bayesian models in project management. Figure 13 (In Appendix) shows the entire violin plot.

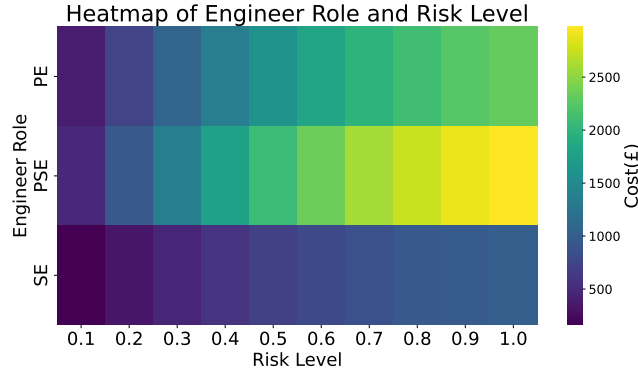


Figure 6: Heat map of each engineering role against the Risk level. The colour represents the value of the interquartile range.

Figure 6 shows that PSEs incur the highest costs, reflecting their higher base rate and variability. However, PEs have higher costs than SEs despite a lower base rate. This may be due to the higher frequency of PE roles in the dataset, amplifying cost variations, or because PE tasks are assigned higher initial costs that, when adjusted for salary variations and risk levels, result in higher overall costs. Additionally, PE tasks might be linked to higher risk levels, suggesting that risk exposure is not evenly distributed across roles. At higher risk levels (0.7–1.0), PSEs still show the widest cost range, but the unexpected cost inversion between PEs and SEs at moderate risk levels indicates the need for model refinements. This could involve adjusting risk assignments or re-evaluating task cost distributions to align better with salary structures. The heat map highlights the importance of validating cost hierarchies against real project data to avoid distorting role-based cost relationships.

5 Dynamic Bayesian Network Model

The unique distinction between Dynamic Bayesian Networks (DBNs) and other traditional BNs is the variance with respect to time [20]. However, the current implementation can be classified

as a quasi-DBN because task dependencies are treated as time-steps rather than representing a true temporal progression as in traditional DBNs. This implementation dynamically updates task distributions by factoring in dependency risks and retrospectively integrating revised risk levels. The risk in this model serves two linked purposes: to vary the spread and centre of the task cost distribution. A higher risk level will provide a wider distribution and a positively shifted centre for cost approximation.

While financial outcomes often define a project’s success, meeting deadlines is also important. Delays can escalate costs, particularly in industries such as construction, where Mckinsey estimates that project overruns cost UK companies £1.4 billion annually [21]. Beyond direct financial losses, missed deadlines can damage reputations, leading to lost contracts. In software development, failing to meet launch schedules can result in missed market opportunities and weakened competitive advantage [22]. Using the classical DBN, analysis can be applied to project timelines alongside costs to provide deeper insights into the influence of risk.

5.1 Method

The DBN model behaves similarly to the models discussed previously. A log-normal cost distribution is generated for each project task but the DBN varies this distribution based on the risk. 10,000 samples are used and the distributions are then further altered depending on the task’s dependencies. For each task, the DBN looks at the predecessor tasks outlined by Synoptix, calculates additional risk based on the combined predecessor risk and then uses a diminishing factor of 0.3 [23] to prevent excessive risk cascade in a scenario with many dependencies. The risk is also further capped at 1 to prevent extreme variations. The equations for how the risk level affects the distribution are as follows:

$$S = S_b + \frac{3(R + R_d)}{10}, \quad L = C_b \times \frac{(1 + R + R_d)}{5}, \quad (2)$$

where S is the scale factor for the new cost distribution, S_b is the base scale for all task distributions, L is the mean location after the shift, R_d is the risk determined from the dependant tasks, and R and C_b are the risk level and base cost preassigned by the contractor respectively.

5.2 Results

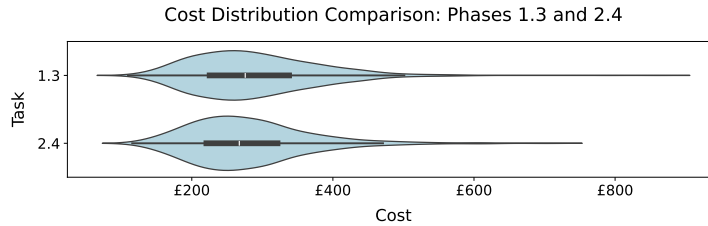


Figure 7: A subsection of the violin plot outlining the distribution of individual task costs within a project phase using the quasi-DBN model. The black box-plot indicates the optimistic and pessimistic cost. The full task violin distribution, 12, is located in the appendix.

Based on the violin plot in Figure 7, tasks with higher dependencies are more likely to result in higher costs under the quasi-DBN model. For example, Tasks 1.3 and 2.4 share a base cost $C_b = 240$ and an initial risk level of 0.4, but Task 1.3, with more dependencies, has an upper bound cost of approximately £339 compared to £322 for Task 2.4. The distribution for Task

1.3 is also more positively skewed, reflecting greater uncertainty due to its dependency on Tasks 1.2 and 1.2.1. This suggests a higher likelihood of exceeding the budget for highly dependent tasks, validating the model’s ability to realistically capture cumulative cost risks.

Figure 8 illustrates the impact of dependency risk on task-level risk severity. Tasks with more dependencies exhibit diminishing returns as risk levels increase, due to a cap on maximum risk. For example, Task 1.1, with no dependencies, shows a linear risk increase, while Task 1.4, dependent on Tasks 1.3, 1.3.1, and 1.3.2, reaches a terminal risk point beyond a level of 0.7, where further increases have minimal effect. This behaviour, unique to the DBN, ensures a more realistic and constrained evaluation of risk escalation compared to traditional methods.

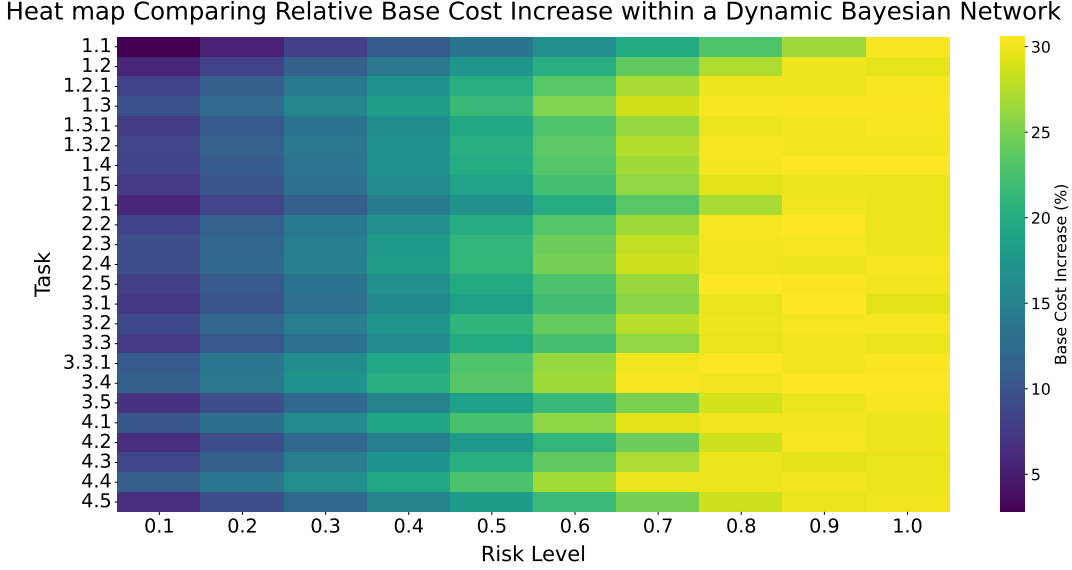


Figure 8: A heat map visualising the relationship between risk level increments and percentage increase from the base cost C_b across different project tasks.

The quasi-DBN provides an improved framework for cost analysis by explicitly capturing temporal dependencies and inter-task relationships, offering a more accurate representation of project progression. However, its main limitation is its inability to account for concurrent task interactions, potentially oversimplifying simultaneous temporal relationships.

Unlike the quasi-DBN, the classical DBN offers a more comprehensive framework by distinguishing between concurrent and sequential dependencies. The Bayesian network graph for Phase 1 (Appendix 14) illustrates this structure, where concurrent tasks proceed simultaneously, while sequential tasks require prior completion of dependencies. Phase 2 cannot begin until all tasks in Phase 1 are finished.

Figure 9 visualises task duration variance using violin plots truncated at the 25th and 75th percentiles to minimise the influence of extreme outliers [24]. Longer tasks exhibit broader distributions, reflecting greater complexity and delay susceptibility. However, risk levels play a more significant role in intra-task variance. For instance, despite Task 4.4 having a longer mean duration, Task 4.3 shows greater variance due to its higher risk level. This underscores the critical role of risk levels in predicting task delays, often outweighing task duration in impact.

While task-level analysis provides valuable insights, it does not fully capture how risks propagate

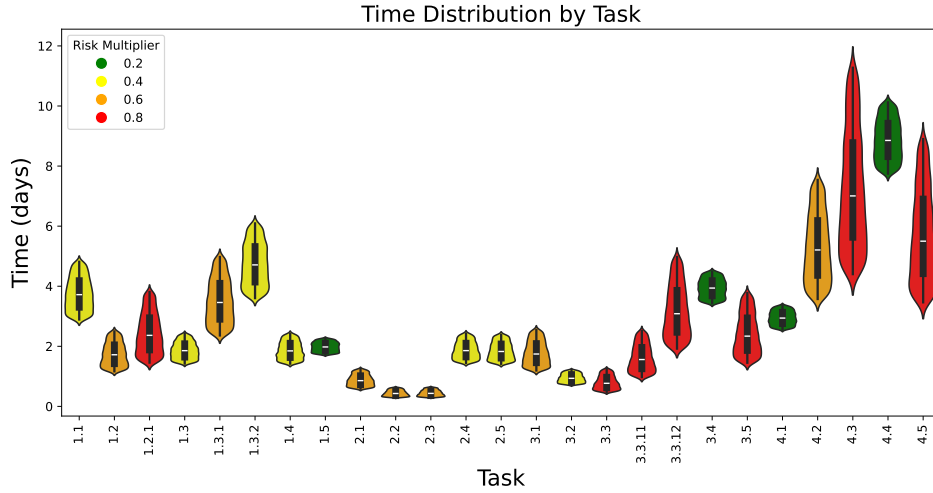


Figure 9: A violin plot showing the distribution of time taken to complete each task in the project, with the data truncated at the 25th and 75th percentiles of the generated log-normal distribution. The colours of the violins correspond to the initial risk level assigned to each task.

across the entire project timeline. A more comprehensive approach considers total project duration and the influence of task-level uncertainties. Figure 10 presents heat maps showing optimistic and pessimistic projections of total project time under varying risk levels. The optimistic estimate corresponds to the upper quartile of the log-normal distribution, while the pessimistic estimate reflects the lower quartile.

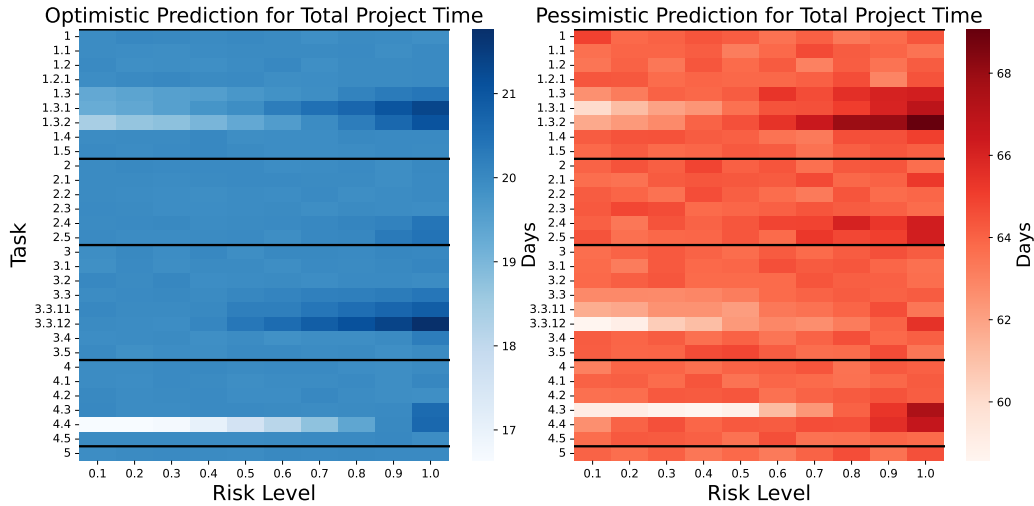


Figure 10: This figure illustrates the projected changes in total project time under optimistic (left) and pessimistic (right) scenarios. The optimistic estimate corresponds to the lower quartile of the log-normal distribution, while the pessimistic estimate reflects the upper quartile. The black lines denote the phase boundaries.

These heat maps reveal that tasks critical to the project timeline are not necessarily those with the longest individual durations but those forming the longest path through the dependency network. For example, Task 1.3.2 significantly affects total project time, particularly in pes-

simistic scenarios. In the optimistic projection, tasks outside the longest path have minimal impact on total duration, even at high-risk levels. The pessimistic projection shows greater variability, with additional tasks influencing duration at lower risk levels. This is attributed to the log-normal distribution, where its right-skewed nature amplifies delays more than time reductions.

Although less directly linked to cost, this analysis highlights phases requiring the most attention in industries where strict deadlines are crucial. Addressing critical tasks is complex and necessitates industry-specific strategies. While increasing resources may seem beneficial, trade-offs such as cost implications must be considered. Moreover, adding personnel does not always enhance efficiency and can lead to diminishing returns [25].

6 Discussion

This paper explores probabilistic techniques for estimating project costs under uncertainty, focusing on Bayesian Networks (BNs) and Monte Carlo Simulations (MCSs). These models capture uncertainties by incorporating probabilistic dependencies and risk factors. While one model may seem optimal for a given application, alternatives can remain viable depending on the context. The study evaluates these models in terms of cost variability, computational complexity, and practical applicability, aiming to balance accuracy and efficiency.

Model	Optimistic (£)	Mean Cost (£)	Pessimistic (£)	Computation Time (ms)
Synoptix Model	6,691.50	7,435.00	8,178.50	N/A
MCS	4,828.10	6,514.90	8,971.30	670
SBN	8,264.00	8,878.00	9,304.00	631
BHM	5,324.10	6,283.60	8,711.77	1,252
DBN	7,956.76	8,828.63	9,638.51	831

Table 2: Key outputs of each modelling approach.

Computational efficiency is crucial when selecting a probabilistic model. However, Table 2 shows that all models exhibit similar computation times, indicating that runtime is more dependent on project complexity rather than the choice of model. As project scale increases, computational demands are expected to rise accordingly [26], where a difference in computation time would then become measurable.

MCSs output the lowest estimation for optimistic costs. This is because they do not take into account dependencies like the BN models which further skew the distribution to the pessimistic side, which is a limitation. Bayesian networks excel at modelling these dependencies, offering a structured approach to probabilistic reasoning [27], while still quantify uncertainty like the MCS. Therefore, MCS would be preferred when dealing with many risk interactions and when a broad exploration of possible cost outcomes is desired [28].

SBNs produce stepwise approximations due to their discrete risk states, limiting their realism. Violin plots reveal that SBNs 16 (in Appendix) lack smooth cost variations, making them unsuitable for detailed cost uncertainty analysis. Additionally, SBNs rely heavily on subjective risk inputs, increasing bias risk. These models are best suited for scenarios with limited data and unknown dependencies, which are uncommon in structured project planning.

BHMs 13 and DBNs 12 generate more realistic, continuous cost distributions. Their violin plots are similar, making accuracy comparisons subjective. This is due to the distributions on different

variables exhibited in the BHMs often having little effect on the overall cost as mentioned in 6. However, the BHMs have a greater concentration of lower-cost estimates, resulting in a more pronounced distribution at the lower end of the curve and introducing slightly more positive skew. Overall, the unique characteristics of BHMs are not efficiently utilised for the current purposes of this project. Given their higher computational complexity, DBNs are the more practical choice for all cost modelling purposes.

The dataset provided by Synoptix represents a short-term, relatively simple project; therefore, all models assume inflation to be negligible. However, for projects extending beyond a year, inflation must be accounted for to ensure cost accuracy [29]. For example, the US Federal Highway Administration adjusts cost estimates to year-of-expenditure dollars by applying an annual inflation rate to the construction midpoint, ensuring financial viability [30]. To incorporate inflation, BNs would treat it as a dependency, while the MCSs would model it as a variable. Among the BN models, the BHM would be the most adaptable, as it already accounts for variable distributions, allowing inflation to be integrated as an additional layer.

Beyond cost estimation, DBNs also model risk impact on project duration. However, as shown in Figure 9, risk levels significantly affect variance. The dataset provided by Synoptix, subjectively assigns the risk level to each task, making it prone to bias [31]. Unique projects usually lack sufficient historical data for a purely data approach to risk allocation, however, there is a precedent of machine learning techniques such as neural networks being used to assign risk levels without human bias [32]. Future research should explore how incorporating machine learning-based risk classification could improve estimations, enabling more dynamic risk assessments that adapt as new data becomes available.

7 Conclusion

This study explored probabilistic methods for project cost estimation, comparing Monte Carlo Simulations (MCS), Static Bayesian Networks (SBN), Bayesian Hierarchical Models (BHM), and Dynamic Bayesian Networks (DBN). The results demonstrated that DBNs offer great versatility in capturing interdependencies and temporal dynamics, making them highly suited for complex, multi-phase projects. Their ability to incorporate dependency risks ensures more realistic and constrained evaluations of cost and risk escalation compared to other methods. In contrast, the MCS model provided computational simplicity and effective cost variability quantification but lacked the capacity to model task dependencies, limiting its utility in interconnected project settings. While the BHM provided valuable multilevel insights, the added computational complexity did not provide significant estimation improvements in this context. SBNs, by contrast, demonstrated utility for early-stage risk planning but lacked the capacity for detailed, dynamic risk modelling. This work provides significant value to Synoptix and other similar firms by demonstrating the suitability of DBNs for realistic project cost estimation, where their methods at the moment are still outdated. The models developed offer actionable insights for risk prioritisation, enabling better informed decision-making for complex projects. Future work should focus on addressing current limitations by integrating inflation modelling to improve long-term cost estimations, especially for extended project timelines. Using machine learning to optimise risk level assignment could further improve predictions and offer a scalable solution for larger, more dynamic projects - especially for BHMs that can better facilitate an introduction of another dependency. These advancements would enhance the applicability and value of the models, offering potential for improving project outcomes across other industries.

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8 Appendix

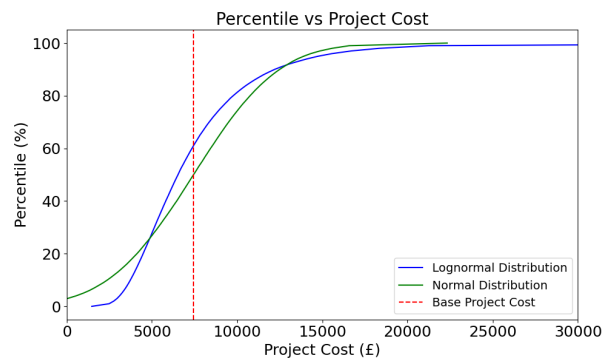


Figure 11

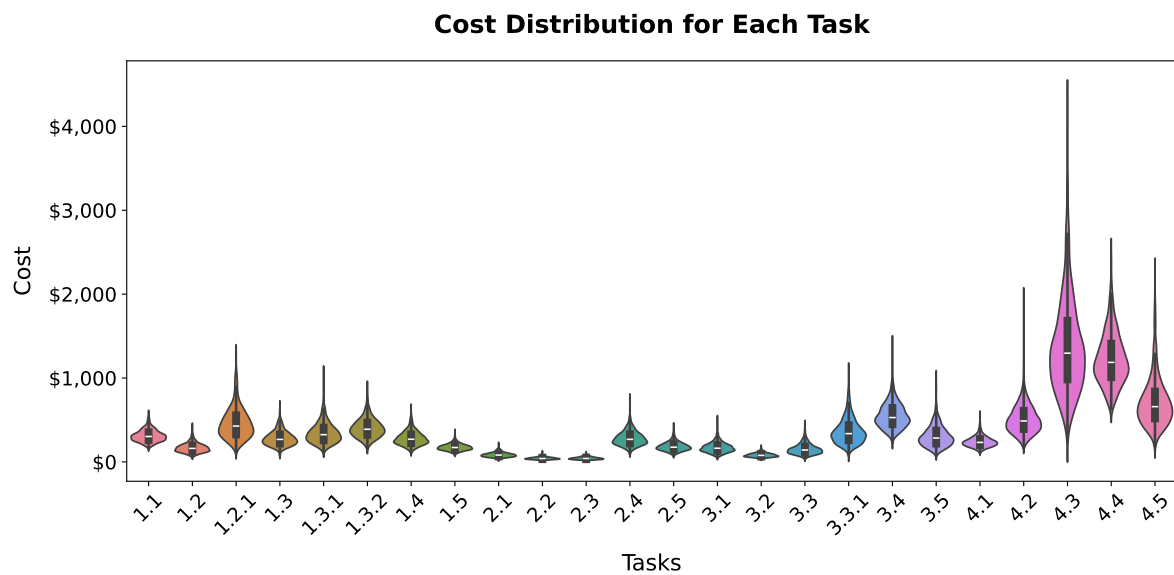


Figure 12: Violin plots outlining the distribution of individual project phase costs using the DBN model

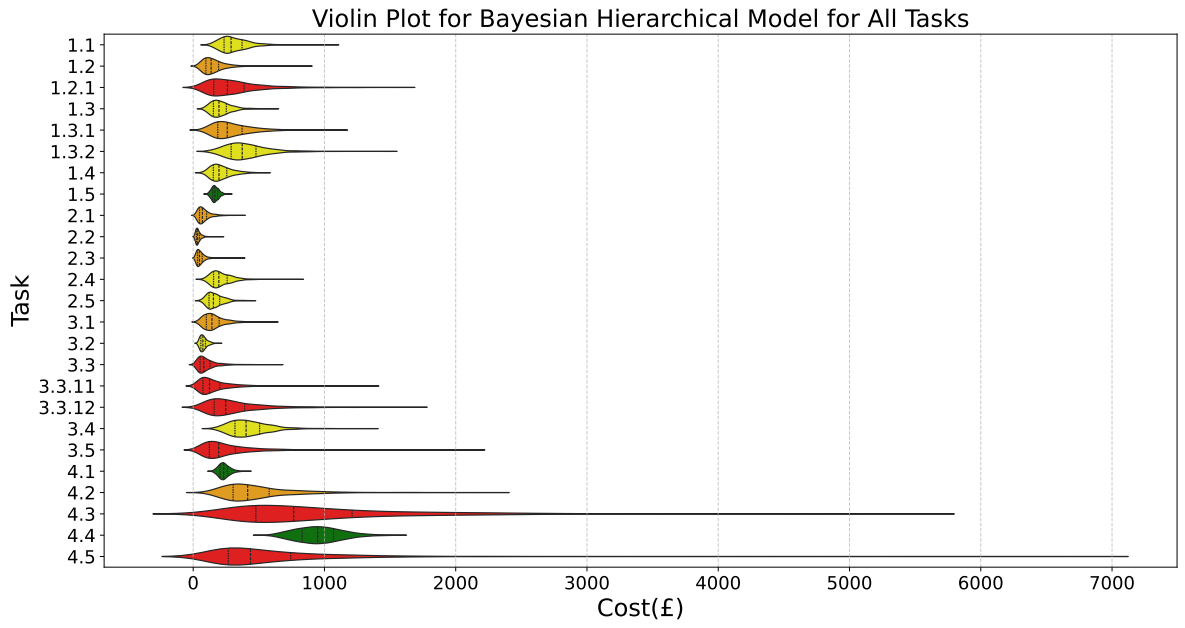


Figure 13: Violin Plot for all the tasks using the Bayesian Hierarchical Model. The colours for each distribution denote the risk category. The left and right lines represent the optimistic and pessimistic costs, respectively and the middle line shows the T_a

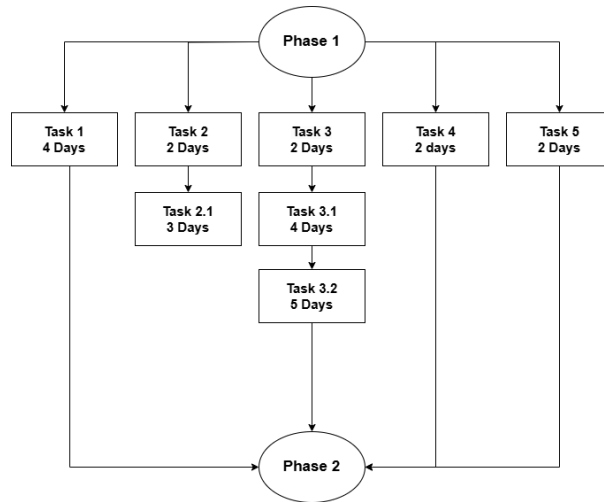


Figure 14: The figure displays a section of the Bayesian network, from Phase 1 to Phase 2. Parallel phases are completed concurrently. The direction of the arrows indicates which phases depend on the completion of others.

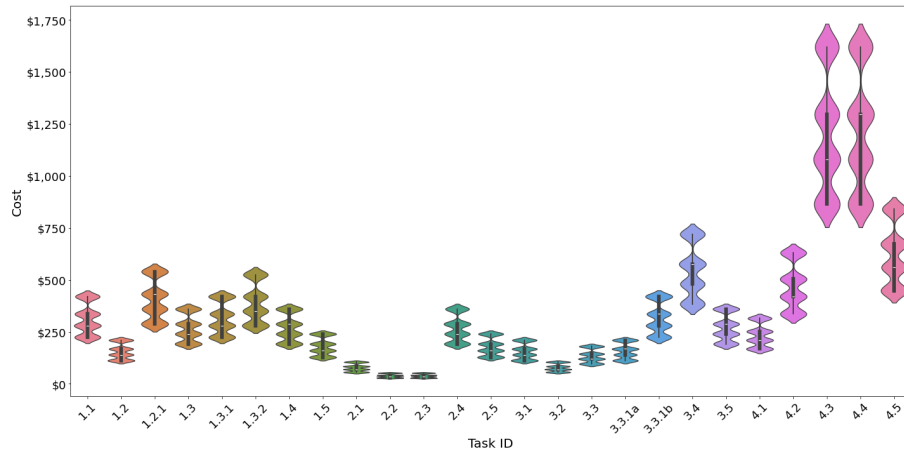


Figure 16: The figure displays violin plots for all tasks using the SBN. The colours used are arbitrary.

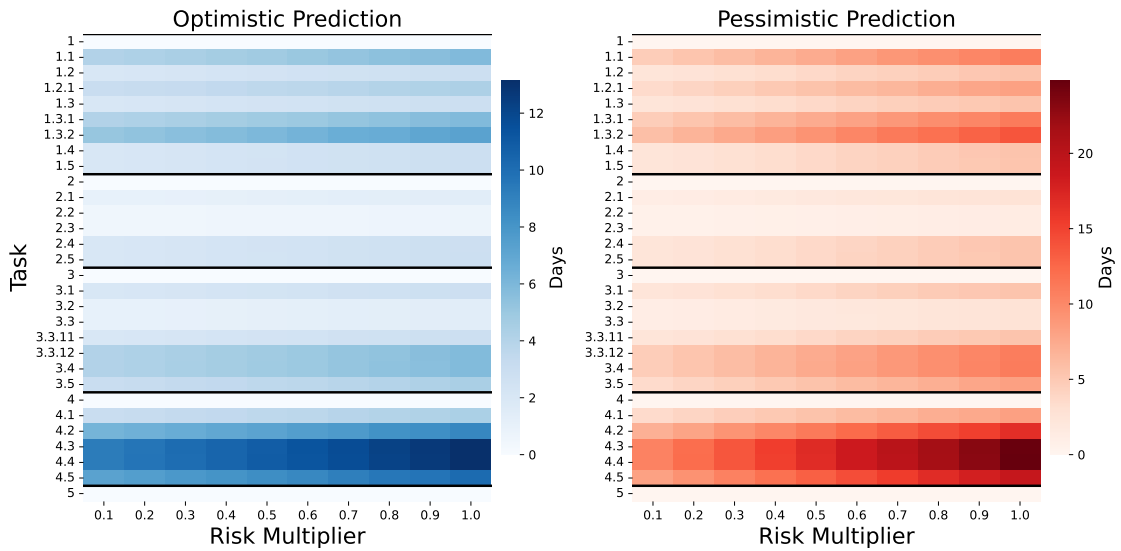


Figure 15: The figure illustrates how the time to complete each phase varies as the risk multiplier in the log-normal distribution changes. On the left-hand side is the optimistic projection, derived from data collected from the lower quartile of the distribution. On the right-hand side is the pessimistic projection, based on the upper quartile. The black lines are used to indicate which phase of the project the tasks belong to.